

Charge-to-mass ratio of an electron

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Abstract

1 Introduction

Historically, J.J. Thompson first determined the charge-to-mass ratio of an electron, a fundamental particle with fixed proportions, which contributed to the development of the atomic model [1], this ratio can be expressed as:

$$\frac{e}{m} \quad (1)$$

In this modified experiment, electrons from an electron gun will be accelerated by a potential difference and passed through a pair of Helmholtz Coils charged by a current, a resultant curved orbit can then be observed in a glass bulb. By considering the relationship between the curvature of voltage-accelerated electron orbits, and the strength of the magnetic field, the following relation can be derived [2]:

$$\frac{1}{r} = \sqrt{\frac{e}{2m}} \frac{B}{\sqrt{\Delta V}} \quad (2)$$

Where r is the radius of the electron orbit, e is the charge of an electron, m is the mass of an electron, B is the magnitude of the generated magnetic field, ΔV is the potential difference accelerating the electrons. The equation that describes the strength of the magnetic field generated by a pair of Helmholtz Coils with equal separation distances, radius and charged with a constant current can be given as [2]:

$$B_c = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 n I}{R} \quad (3)$$

Where B_c is the radius of the electron orbit, R is the separation distance or radius, $\mu_0 = 4\pi \cdot 10^{-7} \text{ T m A}^{-1}$ is the free space permeability, n is the number of turns in the Helmholtz coil, I is the current passed through. The total magnetic field is given by the sum of individual fields, and is given by [2]:

$$B = B_c + B_e \quad (4)$$

Where B_e is the magnitude of the external magnetic field generated by the Earth and instruments and devices nearby. Combining Equations 2, 3, 4, the following equation is obtained:

$$\frac{\sqrt{\Delta V}}{r} = \sqrt{\frac{e}{m}} k \left(I + \frac{1}{\sqrt{2}} I_0 \right) \text{ or } \sqrt{\Delta V} = r \sqrt{\frac{e}{m}} k \left(I + \frac{1}{\sqrt{2}} I_0 \right) \quad k = \frac{1}{\sqrt{2}} \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 n}{R}, I_0 = \frac{B_e}{k} \quad (5)$$

Where the constant k depends on coil dimensions, I_0 is proportional to the strength of the external magnetic field. In the experiment, the potential differences accelerating the electron, currents passing through the Helmholtz Coils will each be held constant one at a time, the radius of the curved orbit is then measured in both instances. Correction factors and the surrounding magnetic field will also be determined. Finally, these measurements will be fit into Equation 5 to obtain a slope proportional to

$k \sqrt{\frac{e}{m}}$, effectively measuring the charge-to-mass ratio.

2 Materials and Methods

2.1 Materials

Equipments:

- Electron Gun
- Glass Bulb
- Helmholtz Coils ($n = 130$, $R = 14.80 \pm 0.05\text{cm}$)
- Anode voltage meter (uncertainty $\pm 0.05\text{V}$)
- Coil current meter (uncertainty $\pm 0.0005\text{A}$)
- Power box
- Anode and filament power supply
- Coils power supply
- Rheostat
- Self-illuminated scale and plastic reflector (uncertainty $\pm 0.5\text{mm}$)

2.2 Methods

Each power source and meters are connected to the Power Box on the Helmholtz coil. as shown in Figure 1, these include 3 separate circuits:

1. Ammeter, rheostat and coils power supply in series to the Helmholtz Coils
2. Anode power supply in series with the accelerating anode, voltmeter parallel to the supply.
3. Filament power supply in series with the electron gun filament.

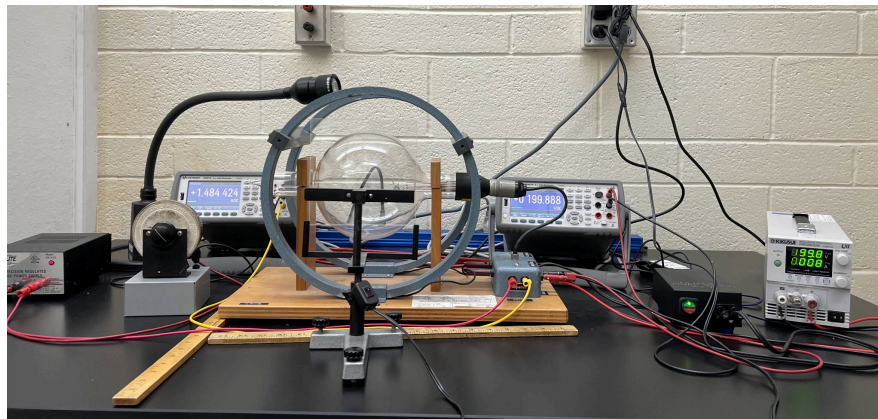


Figure 1: setup of the experiment

The power supplies are then switched on in the order of the filament, anode then the Helmholtz Coils. *Ensure the filament power is not off unless the anode voltage is also off.* Close surrounding light sources to observe the beam path, gently rotate the bulb so that the beam follows a circular, closed path (*non helical*). Place the self-illuminated scale in the center of the coils, align the middle with the center of the glass bulb. The precise alignment allows the elimination of parallax by viewing through the reflection, allowing observations to be made relative to the curved surface¹. For the first trail, adjust the rheostat to vary the coil current, while holding the anode voltage constant, record the diameter of the electron path. For the second trail, repeat the recordings, adjust the power supply to vary the anode voltage, while holding the coil current constant.

¹ Question 1

3 Data and Analysis

[3.1] Correction Factor

Due to the electron path being off-centered from the coil center, the actual magnetic field, $B(\rho)$ located at an off-axis distance, ρ is weaker than the magnetic field strength at the coil center, $B(0)$. Assuming a large radii ($\rho \approx r$), a correction factor for $0.2R < \rho < 0.5R$ can be approximated as [2]:

$$f = \frac{B(\rho)}{B(0)} = 1 - \frac{\rho^4}{R^4 \left(0.6583 + 0.29 \frac{\rho^2}{R^2} \right)^2}, \rho = r \quad (6)$$

A table for this correction factor can be found in *Table 2* in the Appendix. In this analysis, a low voltage and high current is selectively neglected, as it produces a magnetic field of high strength. In such a state the trajectory becomes highly helical as shown in Figure 2, and the velocity of the electron parallel to the field vector becomes significant, and as such, not all parts of the trajectory are equally affected. The helical orbit and the decrease in orbital size causes significant error in measurements. In this experiment, this error is reduced by introducing the correction factor, also by avoiding values of a significant correction factor (< 0.95)².

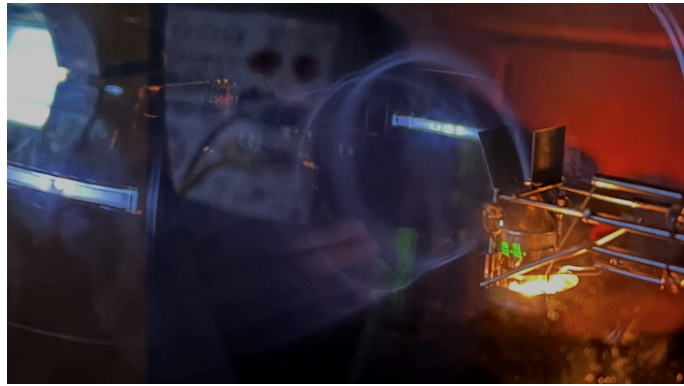


Figure 2: Helical electron beam path at high current and low voltage

[3.2] Ambient Field (B_e)

To determine the ambient fields in the laboratory, (5) can be rewritten as

$$\Rightarrow B_c / f = \alpha \frac{1}{r} - B_e, \text{ where } \alpha = \sqrt{\frac{2m}{e} \Delta V}, \text{ and } B_e \text{ is given by Equation 3} \quad (7)$$

This equation can be modelled as a linear equation of B_c / f as a function of $\frac{1}{r}$ using the data collected at constant voltage (slope is constant, so ΔV must be constant), and the resulting y-intercept is equivalent to the ambient field B_e . A table for the x-values and y-values used to plot this relationship can be found in *Table 3* in the Appendix.

²Question 2

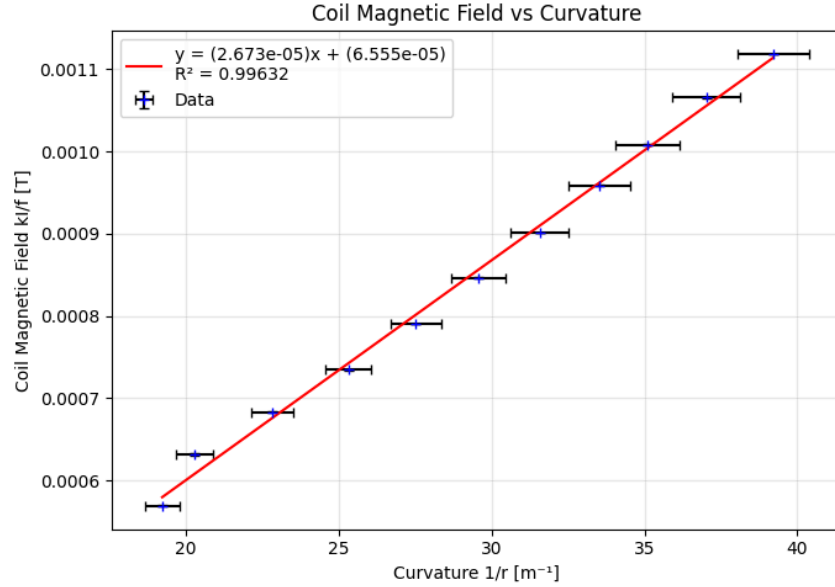


Figure 3, Best fit line of B_c as a function of $1/r$, the R^2 value is 0.9963 and the Reduced Chi Squared value, χ_r^2 is 246.8

In Figure 3, R^2 is very close to 1, indicating a suitable fit, however, $\chi_r^2 \gg 1$, indicating an underestimate in uncertainties, the ambient field obtained, $B_e = (0.7 \pm 0.2) \times 10^{-4}$ T. This value is in good agreement with the accepted literature value of $53018.9 \text{ nT} = 0.53 \times 10^{-4} \text{ T}$ [1] (Earth's magnetic field in Toronto), lying within the experimental uncertainty range, indicating consistency between measured and expected results. The slight positive deviation may result from residual magnetic fields, such as from nearby electronic devices³.

When two iPhones were held tangent to the glass bulb, the electrons were all attracted towards the electronics, and the beam appeared more defined. However, there was no measurable difference in diameter when measuring at constant voltage and current between the presence of electronics, and without. It is important to note that multiple other electronics were in proximity of the apparatus, including two iPads and two laptops, but due to measurement procedure practicality, only the effect of two iPhones could be evaluated. Moreover, there were two other identical apparatuses occurring simultaneously in the same lab space. Therefore, the additional magnetic field generated by these devices has a small influence on the calculated field, explaining the slight deviation from the literature value⁴.

[3.3] Charge-to-Mass Ratio (e/m)

To determine the charge-to-mass ratio (e/m), equation (5) can be modelled as a linear equation of $\frac{\sqrt{\Delta V}}{r}$ as a function of $\left(I + \frac{1}{\sqrt{2}}I_0\right)/f$ when considering the constant potential difference data set. When applying

³Question 3

⁴Question 4

the constant current data set, $\sqrt{\Delta V}$ can be modelled as a function of $\left(I + \frac{1}{\sqrt{2}}I_0\right)r/f$. In both equations,

the slope (β) is equivalent to $\sqrt{\frac{e}{m}}k$.

The charge-to-mass ratio can thus be calculated by:

$$\frac{e}{m} = \left(\frac{\beta}{k}\right)^2, \text{ where } k = 5.58 \cdot 10^{-4} \pm 0.00 \text{ T A}^{-1} \quad (8)$$

A table for the x-values and y-values used to plot this relationship can be found in *Table 4a, 4b* in the Appendix.

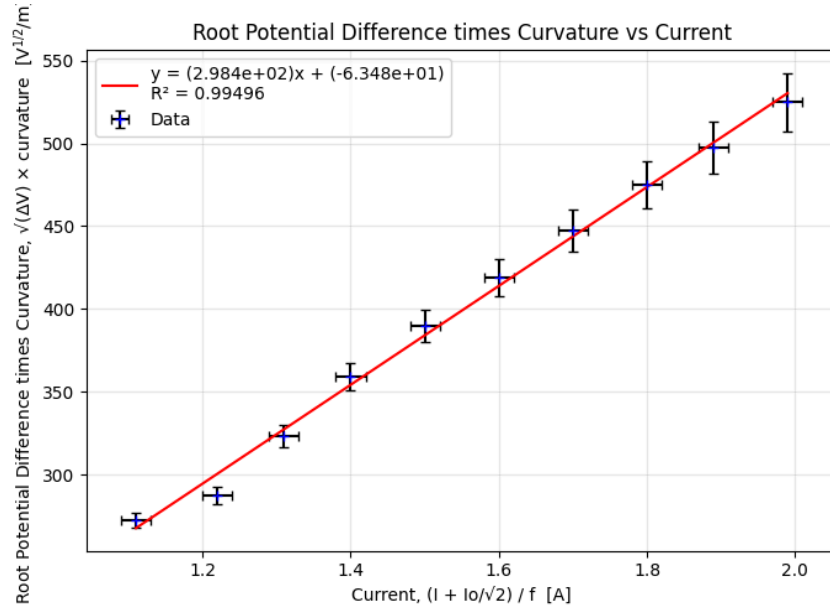


Figure 4, Best fit line for the constant potential difference data set, $\frac{\sqrt{\Delta V}}{r}$ as a function of $\left(I + \frac{1}{\sqrt{2}}I_0\right)/f$, the R^2 value is 0.9949 and the Reduced Chi Squared value, χ_r^2 is 1.118

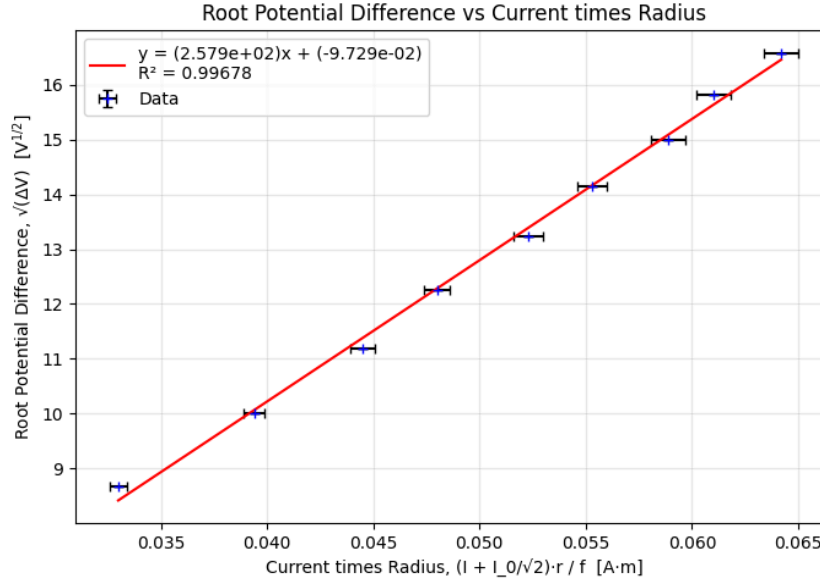


Figure 5, Best fit line for the constant current data set, $\sqrt{\Delta V}$ as a function of $\left(I + \frac{1}{\sqrt{2}}I_0\right)r/f$, the R^2 value is 0.9968 and the Reduced Chi Squared value, χ_r^2 is 5.409×10^7

For the constant potential difference fit, R^2 and χ_r^2 is very close to 1, indicating a suitable fit and estimate in uncertainties, the slope is $(29.7 \pm 0.7) \times 10$, the charge-to-mass ratio is $(28.2 \pm 0.7) \times 10^{10} \text{ C kg}^{-1}$. For the constant current fit, R^2 is very close to 1, indicating a suitable fit, but $\chi_r^2 \gg 1$, this indicates a severe underestimation in uncertainties, the slope is $(25.7 \pm 0.6) \times 10$, the charge-to-mass ratio is $(21.2 \pm 0.5) \times 10^{10} \text{ C kg}^{-1}$. Taking the average of the ratio calculated from the two data sets, the charge-to-mass ratio determined is $(24.8 \pm 0.4) \times 10^{10} \text{ C kg}^{-1}$ ⁵.

4 Discussion and Conclusion

This experiment is a variation of J.J. Thomson's derivation of the electron's charge-to-mass ratio, based on measuring the deflection of a charged particle moving through a magnetic field. Measurements were conducted by incrementally varying the potential difference and current to evaluate both the influence and magnitude of the ambient magnetic field, as well as to determine the charge-to-mass ratio, incorporating a correction factor to account for the influence of off-axis distance on the generated field. Plotting the potential difference as a function of the total magnetic field, the slope is equivalent to the square root of the electron charge-to-mass ratio, found to be $(24.8 \pm 0.4) \times 10^{10} \text{ C kg}^{-1}$.

The literature value for the charge-to-mass ratio of the electron is $1.758820 \times 10^{11} \text{ C kg}^{-1} \approx 17.6 \times 10^{10} \text{ C kg}^{-1}$ [4], which lies 18 times outside the experimental value's uncertainty, implying high imprecision. Additionally, $\chi_r^2 \gg 1$ (5.409×10^7) when using the constant current data set, suggesting uncertainties are significantly underestimated. Moreover, the experimental value demonstrates low accuracy as it deviates by 41.0% from the theoretical value. Despite this discrepancy, the result is of the correct order of magnitude and demonstrates reasonable agreement with theoretical expectations.

⁵Question 5

Two potential sources of error are significant to this experiment. Firstly, while performing the experiment, it was noted that the voltage accumulates as the experiment progresses, causing constant fluctuations in both voltage and current readings. Fluctuations in accelerating voltage and coil current cause inconsistent beam velocities and field strengths. This accentuates the error in the final result as the potential difference readings were used to calculate the ambient magnetic field – an essential variable for the final ratio calculation. Even small errors in the magnetic field value result in large effects on the charge-to-mass ratio as it is proportional to $\frac{1}{B^2}$. Secondly, the electron beam was not perfectly centered along the coil axis, causing the electrons to experience a non-uniform or weaker field. This off-axis trajectory yields a smaller radius of curvature than predicted, influencing the correction factor and thus ratio value.

In conclusion, the charge-to-mass ratio is experimentally derived as $(24.8 \pm 0.4) \times 10^{10} \text{ C kg}^{-1}$, which deviates from theory by 41.0% and outside its uncertainty range. Therefore, this experiment was not precise or accurate to determine the charge-to-mass ratio of the electron.

Reference

- [1] H. Zhan, E. Horsley, and A. Harlick, *Charge-to-mass ratio for the electron*, University of Toronto, Revision 3fcb776, Mar. 17, 2025. [Online]. Available: https://www.physics.utoronto.ca/~phy224_324/LabManuals/ChargeToMassRatio.pdf
- [2] “Historical Development of Atomic Theory,” *Inorganic Chemistry (LibreTexts)*. [Online]. Available: [https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Inorganic_Chemistry_\(LibreTexts\)/02%3A_Atomic_Structure/2.01%3A_Historical_Development_of_Atomic_Theory](https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Inorganic_Chemistry_(LibreTexts)/02%3A_Atomic_Structure/2.01%3A_Historical_Development_of_Atomic_Theory). [Accessed: Oct. 13, 2025].
- [3] “Magnetic declination in Toronto, Canada: -10.14° ($-10^\circ 8'$),” *Magnetic-Declination.com*, 2025.
- [4] “The CODATA recommended value for the electron charge-to-mass quotient is $-1.758\,820\,008\,38(55) \times 10^{11} \text{ C} \cdot \text{kg}^{-1}$,” *Wikipedia: Mass-to-charge ratio*, accessed Oct. 2025. en.wikipedia.org

Sample Calculations

Uncertainty of k can be given as:

$$\frac{\Delta k}{k} = \frac{1}{\sqrt{2}} \left(\frac{4}{5} \right)^{3/2} \mu_0 n \cdot \frac{\Delta R}{R} \quad (\text{a})$$

Uncertainty of averaging radius over 3 trials can be given as:

$$\Delta r_{avg} = \sqrt{3\Delta r^2} = \sqrt{3}\Delta r \quad (\text{b})$$

Uncertainty of correction factor can be given as:

$$\Delta \frac{B(\rho)}{B(0)} = \sqrt{\left(\frac{\partial f}{\partial \rho} \Delta \rho \right)^2 + \left(\frac{\partial f}{\partial R} \Delta R \right)^2} = \sqrt{\left(\frac{2.6332R^2 \rho^3}{(0.6583R^2 + 0.29\rho^2)^3} \Delta \rho \right)^2 + \left(\frac{2.6332R\rho^4}{(0.6583R^2 + 0.29\rho^2)^3} \Delta R \right)^2} \quad (\text{c})$$

Uncertainty of $\frac{1}{r}$ can be given as:

$$\Delta \left(\frac{1}{r} \right) = \frac{\Delta r}{r^2} \quad (\text{d})$$

Uncertainty of B_c can be given as:

$$\Delta B_c = B_c \sqrt{(kI\Delta f)^2 + (fI\Delta k)^2 + (fk\Delta I)^2} \quad (\text{e})$$

Uncertainty of I_0 can be given as:

$$\Delta I_0 = \sqrt{\left(\frac{\Delta k}{k} \right)^2 + \left(\frac{\Delta B_e}{B_e} \right)} \quad (\text{f})$$

Uncertainty of $\sqrt{\Delta V}$ can be given as:

$$\Delta(\sqrt{\Delta V}) = \frac{1}{2} \sqrt{\Delta V} \cdot \frac{\Delta(\Delta V)}{\Delta V} \quad (\text{g})$$

Uncertainty of $\frac{(I + \frac{1}{\sqrt{2}}I_0)r}{f}$ can be given as:

$$\text{Let } g = \left(I + \frac{1}{\sqrt{2}}I_0\right), x = \frac{(I + \frac{1}{\sqrt{2}}I_0)r}{f} = \frac{gr}{f}$$

$$\Delta x = \sqrt{\left(\frac{g\Delta r}{f}\right)^2 + \frac{r^2}{f^2}((\Delta I)^2 + \frac{(\Delta I_0)^2}{2}) + \left(\frac{x\Delta f}{f}\right)^2} \quad (\text{h})$$

Uncertainty of $\frac{\sqrt{\Delta V}}{r}$ can be given as:

$$\Delta\left(\frac{\sqrt{\Delta V}}{r}\right) = \frac{\sqrt{\Delta V}}{r} \sqrt{\left(\frac{\sigma_{\sqrt{\Delta V}}}{\sqrt{\Delta V}}\right)^2 + \left(\frac{\Delta r}{r}\right)^2} \quad (\text{i})$$

Uncertainty of $\frac{(I + \frac{1}{\sqrt{2}}I_0)}{f}$ can be given as:

$$\text{Let } g = \left(I + \frac{1}{\sqrt{2}}I_0\right), x = \frac{(I + \frac{1}{\sqrt{2}}I_0)}{f} = \frac{g}{f}$$

$$\Delta x = \sqrt{\frac{(\Delta I)^2 + \frac{1}{2}(\Delta I_0)^2}{f^2} + \left(\frac{x\Delta f}{f}\right)^2} \quad (\text{j})$$

Uncertainty of charge-to-mass ratio can be given as:

$$\Delta\left(\frac{e}{m}\right) = \frac{e}{m} \sqrt{\left(\frac{\Delta\beta}{\beta}\right)^2 + \left(\frac{\Delta k}{k}\right)^2} \quad (\text{k})$$

Uncertainty of averaging charge-to-mass ratio between two plots:

$$\Delta\left(\frac{e}{m}\right)_{avg} = \frac{1}{2} \sqrt{\left(\Delta\left(\frac{e}{m}\right)_{const V}\right)^2 + \left(\Delta\left(\frac{e}{m}\right)_{const I}\right)^2} \quad (\text{l})$$

Reduced Chi squared: (m)

$$\chi_r^2 = \frac{1}{N-m} \sum_{i=1}^N \frac{[y_i - f(x_i)]^2}{\sigma_{yi}^2}, \text{ N is the number of data points, m is the fit parameter (2), } \sigma_{yi}^2 \text{ is the uncertainty, } y_i - f(x_i) \text{ is the fit residual.}$$

Additional Data Tables

Constant	Value
Free Space Permeability, μ_0	$4\pi \cdot 10^{-7} \text{ T m A}^{-1}$

Radius, Separation Distance, R	$14.80 \pm 0.05\text{cm}$ or $0.1520 \pm 5 \cdot 10^{-4}\text{m}$
Number of turns in Helmholtz Coil, n	130
Coil Dimension constant, k	$5.58 \cdot 10^{-4} \pm 0.00\text{ T A}^{-1}$

Table 1: Constants used in the analysis.

Off-axis distance, ρ [m]	Correction Factor $B(\rho)/B(0)$
Constant Voltage Trial	
0.0152	0.9997 ± 0.0001
0.0208	0.9991 ± 0.002
0.0248	0.9982 ± 0.0003
0.0280	0.9971 ± 0.0004
0.0302	0.9962 ± 0.0005
0.0328	0.9946 ± 0.0006
0.0347	0.9934 ± 0.0007
0.0368	0.9916 ± 0.0008
0.0382	0.9904 ± 0.0009
0.0398	0.9886 ± 0.0010
Constant Current Trial	
0.0255	0.9980 ± 0.0003
0.0270	0.9975 ± 0.0003
0.0285	0.9969 ± 0.0004
0.0298	0.9963 ± 0.0004
0.0317	0.9954 ± 0.0005
0.0338	0.9940 ± 0.0006
0.0363	0.9920 ± 0.0008

0.0395	0.9890 ± 0.0010
0.0438	0.9835 ± 0.0013
0.0493	0.9741 ± 0.0018
0.0520	0.9684 ± 0.0021
Phone Trial	
0.0370	0.9915 ± 0.0008

Table 2: Correction Factor for off-axis distance ρ

x-value: $1/r$ [m^{-1}]	x-uncertainty [m^{-1}]	y-value: B_c [T]	y-uncertainty [T]
39.2	1.4	0.0011184	0.0000004
37.0	1.2	0.0010653	0.0000004
35.1	1.1	0.0010078	0.0000005
33.5	1.0	0.0009587	0.0000005
31.6	0.9	0.0009013	0.0000005
29.6	0.8	0.0008464	0.0000006
27.5	0.7	0.0007904	0.0000007
25.3	0.6	0.0007343	0.0000008
22.8	0.5	0.0006831	0.0000009
20.3	0.4	0.0006312	0.0000012
19.2	0.3	0.0005694	0.0000012

Table 3: B_c (field by coils) for $1/r$ (reciprocal of radius)

x-value: $\left(I + \frac{1}{\sqrt{2}}I_0\right)/f$	x-uncertainty	y-value: $\frac{\sqrt{\Delta V}}{r}$	y-uncertainty
1.99	0.02	524.9600	17.4987
1.89	0.02	497.3305	15.7052
1.80	0.02	475.1035	14.3327
1.70	0.02	447.5975	12.7212
1.60	0.02	418.9336	11.1441
1.50	0.02	390.1079	9.6632
1.40	0.02	358.8334	8.1760

1.31	0.02	323.3594	6.6393
1.22	0.02	287.3092	5.2415
1.11	0.02	272.5754	4.7177

Table 4a: plotted data using constant voltage data set, $\frac{\sqrt{\Delta V}}{r}$ for $\left(I + \frac{1}{\sqrt{2}}I_0\right)/f$

x-value: $\left(I + \frac{1}{\sqrt{2}}I_0\right)r/f$	x-uncertainty	y-value: $\sqrt{\Delta V}$	y-uncertainty
0.0330	0.0004	8.66218	0.00003
0.0394	0.0005	10.00333	0.00002
0.0445	0.0006	11.18332	0.00002
0.0480	0.0006	12.25017	0.00002
0.0523	0.0007	13.23128	0.00002
0.0553	0.0007	14.14449	0.00002
0.0589	0.0008	15.00222	0.00002
0.0610	0.0008	15.81350	0.00002
0.0642	0.0008	16.58212	0.00002

Table 4b: plotted data using constant current data set, $\sqrt{\Delta V}$ for $\left(I + \frac{1}{\sqrt{2}}I_0\right)r/f$